

# Project Design Phase

## Problem – Solution Fit Template

|               |                                 |
|---------------|---------------------------------|
| Date          |                                 |
| Team ID       |                                 |
| Project Name  | Credit Card Approval Prediction |
| Maximum marks |                                 |

### Problem – Solution Fit Template

| Problem                                                 | Current Challenges                                                      | Proposed Solution                                                                      | Benefits                                |
|---------------------------------------------------------|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------|-----------------------------------------|
| Manual credit card approval is slow and time-consuming. | Bank staff spend significant time reviewing every application manually. | Use a Machine Learning model to automatically predict credit card approval.            | Faster processing and reduced workload. |
| Human errors in approval decisions.                     | Manual verification may lead to inconsistent decisions.                 | Train the model using historical applicant and credit data for consistent predictions. | Improved accuracy and reliability.      |
| Large number of credit card applications.               | High application volume delays approval.                                | Automate the approval process using the prediction system.                             | Quick response to customers.            |
| Difficulty identifying eligible applicants.             | Banks must analyze multiple applicant attributes.                       | Use features such as income, employment, education, and credit history for prediction. | Better applicant evaluation.            |
| Poor customer experience due to delays.                 | Customers wait a long time for approval results.                        | Provide instant prediction through a Streamlit web application.                        | Improved customer satis                 |

### Problem–Solution Fit Summary

#### Problem

- Manual credit card approval process
- Time-consuming verification
- Human errors
- Delayed customer response
- Inconsistent approval decisions



#### Solution

**Credit Card Approval Prediction using Machine Learning**

- Data preprocessing
- Feature engineering
- Random Forest prediction model
- Streamlit web application
- Instant prediction results



### **Benefits**

- Faster approval process
- Improved prediction accuracy
- Reduced manual effort
- Better decision-making
- Enhanced customer experience